Relativistic detuning, electron heating and the reach of continuum kinetic methods in resonant dark-photon conversion in a plasma

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Hook, Huang and Shalaby (HHS) used one-dimensional relativistic particle-in-cell (PIC) simulations to argue that nonlinear plasma dynamics cuts off the resonant conversion of dark-photon dark matter into plasma oscillations. We examine their resonant case ($v_q/v_{te} = 0.03$) with a noise-free Hermite–Fourier Vlasov solver (nonrelativistic), a relativistic continuum grid solver, a relativistic momentum-space Hermite solver, and an independent relativistic PIC code. Three results follow. (i) Relativistic detuning, absent from nonrelativistic models, sets the collapse of the mean-field response: in both relativistic continuum models the work $W_{\rm ext}$ leaves the linear resonant law by 10 % at $\omega_{pe}t \simeq 392.5$ for $v_q/v_{te} = 0.03$ and at $\simeq 1309.5$–1413.5 for 10^{-3}, whereas nonrelativistic models give 769.5–866.5 or no collapse; the mean-field energy is 0.041 of the linear value at $\omega_{pe}t = 1000$ in both. (ii) The electron heating of HHS, re-digitized from their figure (50.1 initial thermal energies $E_{\rm th}$ at $\omega_{pe}t = 1000$), is reproduced to a factor ~ 2: the relativistic PIC converges to a plateau of $\simeq 25\,E_{\rm th}$ (changes ≤ 1.4 % over 412–1648 particles per cell and for cells of 0.02 to $0.01\,c/\omega_{pe}$), against $62.7 \pm 9.4\,E_{\rm th}$ for a Newtonian pusher, which overestimates the relativistic value by a factor 2.5. The remaining factor between our plateau and HHS is unresolved. (iii) The deterministic continuum codes resolve the detuning but not the heating: they give $0.056\,E_{\rm th}$ at $\omega_{pe}t = 1000$ (certified to $\omega_{pe}t \simeq 1530$), some 445 times below the PIC, and with PIC-like noise they lose positivity at once for the PIC particle number and at $\omega_{pe}t \lesssim 407.5$ for a noise variance 10^3 times lower, before heating develops. Whether finite-amplitude fluctuations absent from the continuum seeds, including cell-scale PIC noise, produce the heating is an open question. We also retain the earlier results of the nonrelativistic solver: its verification, a truncation instability of the asymmetrically weighted Hermite basis that bounds its certified windows to 1000, 892, 791.5 at $v_q/v_{te} = 0.03, 0.05, 0.07$, absence of resolved detuning from a finite reservoir or coherence time, and collisional absorption equal to the linear Drude law; these are statements about a nonrelativistic model whose quiver speed reaches 0.47–0.87 c.

I. INTRODUCTION

A dark photon with kinetic mixing χ with the ordinary photon and a mass $m_{A'}$ is a well-motivated candidate for dark matter [1]. In the early Universe the dark-photon field oscillates coherently, and when $m_{A'}$ equals the plasma frequency of the ionized medium the field resonantly converts into longitudinal plasma oscillations, depositing energy in the plasma [2–4]. Cosmological constraints on dark-photon dark matter have been derived from the resonant transfer of a substantial part of the dark-photon energy density into the plasma, which heats it.

Hook, Huang and Shalaby (HHS) argued that the resonant transfer saturates because of plasma nonlinearities [5]. In their account, large-amplitude $k \simeq 0$ Langmuir waves, driven by the dark photon, excite higher-k Langmuir and ion-acoustic waves through the ponderomotive force; the resulting density inhomogeneities detune the resonance and limit the deposited energy to roughly the electron thermal energy for weak drives, which would weaken the constraints by several orders of magnitude. Their evidence comes from one-dimensional PIC simulations (SHARP, with a relativistic Vay pusher) of an electron–ion plasma with $m_i/m_e = 1836$ and a prescribed uniform dark-photon drive: resonant runs at $v_q^D/v_{th}^e = 0.03$ and 10^{-3} (their Fig. 2) and Landau–Zener runs (their Fig. 3), published as Phys. Rev. Lett. 137, 071004.

This paper asks where, in a model with the same physical ingredients, the energy transfer from a resonant uniform drive departs from the linear resonant law, and what sets the electron heating that HHS plot. We write $v_q/v_{te} \equiv E_0/(v_{te}\omega_{pe})$ for the drive strength, with E_0 the drive amplitude, v_{te} the electron thermal speed and ω_{pe} the electron plasma frequency (Sec. II D). The work proceeded in two stages. First, a noise-free nonrelativistic Hermite–Fourier Vlasov solver, whose seeds are deterministic and whose work ledger closes to solver tolerance, was verified, its failure mode characterized, and used to bound the transfer (Secs. III–VI); its model is nonrelativistic, which turned out to be decisive (the coherent quiver speed it implies is 0.47–0.87 c). Second, relativistic versions of the continuum model (a grid solver and a momentum-space Hermite solver) and an independent relativistic PIC code were run in the HHS set-up (Secs. VII–IX). The central findings are that relativistic detuning explains the timing of the mean-field collapse, that an independent relativistic PIC reproduces the HHS electron heating to a factor of order two while a Newtonian pusher overestimates it, and that the continuum methods resolve the first but not the second. Failed and

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amended checks are reported where they affect a claim (Table II); gates were committed before the runs and not amended.

II. MODEL

A. Canonical Maxwell–Proca–Vlasov system

Each species s obeys the collisionless Vlasov equation in the combined force field,

$$\partial_t f_s + \mathbf{v}\cdot\nabla f_s + \frac{q_s}{m_s}\Big[\mathbf{E} + \eta\mathbf{E}_D + \mathbf{E}_{\rm drive} + \mathbf{v}\times(\mathbf{B} + \eta\mathbf{B}_D)\Big]\cdot\nabla_v f_s = 0, \quad (1)$$

with ordinary fields $\partial_t\mathbf{E} = c^2\nabla\times\mathbf{B} - \mathbf{J}/\epsilon_0$, $\partial_t\mathbf{B} = -\nabla\times\mathbf{E}$ and canonical Proca fields

$$\partial_t\mathbf{E}_D = c^2\nabla\times\mathbf{B}_D + \Omega_D^2\mathbf{A}_D - \eta\mathbf{J}/\epsilon_0, \quad (2)$$
$$\partial_t\mathbf{B}_D = -\nabla\times\mathbf{E}_D, \quad \partial_t\mathbf{A}_D = -\mathbf{E}_D - \nabla\phi_D, \quad (3)$$
$$\partial_t\phi_D = -c^2\nabla\cdot\mathbf{A}_D, \quad (4)$$

with constraints $\nabla\cdot\mathbf{E} = \rho/\epsilon_0$ and $\nabla\cdot\mathbf{E}_D + \Omega_D^2\phi_D/c^2 = \eta\rho/\epsilon_0$. The ordinary current $\mathbf{J}$ sources both laws; the dark field acts back on the particles with strength η. Three modes are used: *ordinary* (no dark coupling), *prescribed drive* (a uniform $\mathbf{E}_{\rm drive} = \mathbf{E}_0\cos(\omega t + \varphi)$ and an external-work ledger, no dark reservoir; the setting of HHS), and *self-consistent* (evolved Proca fields with a finite energy reservoir).

B. Mixing map

For a kinetic mixing $-\frac{\chi}{2}F_{\mu\nu}F_D^{\mu\nu}$ and a mass term on the dark field only, removing the mixing by a non-orthogonal shear gives the canonical coupling

$$\eta = \frac{\chi}{\sqrt{1-\chi^2}}, \qquad m_D = \frac{m_0}{\sqrt{1-\chi^2}}, \quad (5)$$

where m_0 is the bare mass and Ω_D is the dark-photon mass m_D expressed as a frequency. We checked this map against Sec. 1.1 of Ref. [1] (their $\varepsilon \equiv \chi$): the dark photon couples to the ordinary current with strength $e\varepsilon/\sqrt{1-\varepsilon^2}$. The sign convention differs by $\mathbf{A}_D \to -\mathbf{A}_D$, under which the model is symmetric in η. The model is not the massless "millicharged" choice of the same reference.

C. Energy and work ledger

In the parent normalization velocities are in units of c, $\Omega_0 \equiv \Omega_{c,0}$ and the species mass is $m_s = \Omega_0/\Omega_{cs}$, which is required for the energy theorem. With $\langle\cdot\rangle$ the box average the field energies and powers are

$$U_\gamma = \tfrac{1}{2}\Omega_0^2\langle E^2 + B^2\rangle, \quad (6)$$
$$U_D = \tfrac{1}{2}\Omega_0^2\big\langle E_D^2 + B_D^2 + \Omega_D^2(A_D^2 + \phi_D^2)\big\rangle, \quad (7)$$
$$P_\gamma = \Omega_0\langle\mathbf{J}\cdot\mathbf{E}\rangle, \quad P_D = \Omega_0\eta\langle\mathbf{J}\cdot\mathbf{E}_D\rangle, \quad (8)$$
$$P_{\rm ext} = \Omega_0\bar{\mathbf{J}}\cdot\mathbf{E}_{\rm drive}. \quad (9)$$

The works W_γ, W_D, $W_{\rm ext}$ are integrated as extra ODE states with the same Runge–Kutta stages as the fields, so that $K - K_0 = W_\gamma + W_D + W_{\rm ext}$, $U_\gamma - U_{\gamma 0} = -W_\gamma$ and $U_D - U_{D0} = -W_D$ hold to solver tolerance. The Proca energy flux is $\epsilon_0 c^2\mathbf{E}_D\times\mathbf{B}_D + \epsilon_0\Omega_D^2\phi_D\mathbf{A}_D$; omitting the $\phi_D\mathbf{A}_D$ term breaks the flux budget (Sec. IV). For a homogeneous plasma under the uniform drive the mean field obeys a driven oscillator at the plasma frequency $\omega_L = \sqrt{1 + m_e/m_i}\,\omega_{pe}$, whose solution is the *linear resonant law*: $\dot{\bar{E}} = -\bar{J}$, $\dot{\bar{J}} = \omega_L^2(\bar{E} + E_0\cos\omega t)$, $\dot{W} = \bar{J}E_0\cos\omega t$. This law is exact in the homogeneous discrete Hermite system because only Hermite orders 0–1 enter (a homogeneous check, agreement to 10^{-9} for $\omega_{pe}t \le 1000$ even at quiver speeds of $15\,v_{te}$). With fluctuations present it is the baseline against which the transfer is measured, and the departure from it contains the $\langle\delta n\,\delta E\rangle$ back-reaction term of the mean-field identity.

D. Units and the HHS-like set-up

All simulations use $c = \omega_{pe} = n = 1$ for the electrons, $m_i/m_e = 1836$, $T_i = T_e$, $v_{te}^2 = 10^{-3}c^2$, box length $L = 40\,c/\omega_{pe}$ and a prescribed drive at $\omega = \omega_L$ unless stated. We call $v_q/v_{te} \equiv E_0/(v_{te}\omega_{pe})$ the effective initial force (the quiver speed grows secularly with time at resonance). Energies are quoted in units of nT_eV. HHS use $v_q^D/v_{th}^e = 0.03$ and 10^{-3} for their resonant runs (PRL Fig. 2), $m_i/m_e = 1836$, $k_BT_e = k_BT_i = 10^{-3}m_ec^2$ ($v_{th}^e = 3.1\times10^{-2}c$), $L = 40\,c/\omega_{pe}$ with 1000 cells of $0.04\,c/\omega_{pe}$ and a prescribed spatially uniform drive $A_0\cos\omega t$ (Supplemental Material, Sec. II); we use the same v_{te}, mass ratio, temperature ratio, box length and v_q definition, but not the loading (their particles start equally spaced, ions and electrons colocated; ours is a deterministic spectral seed) or the relativistic pusher. The nonrelativistic runs are *HHS-inspired*, not reproductions; the relativistic runs of Secs. VII–IX use the same set-up and, for the PIC, the HHS loading conventions as far as they are stated.

E. Relativistic extensions

Three further models share the set-up of Sec. II D and differ from the nonrelativistic solver only in the particle dynamics. In all three the momentum per unit mass $\mathbf{u} = \gamma\mathbf{v}$ replaces the velocity as the kinetic variable, $\gamma = \sqrt{1 + u^2/c^2}$, the force term is unchanged, and streaming,

the current and the kinetic energy $K = \int(\gamma-1)mc^2 f$ use $\mathbf{v}(\mathbf{u}) = \mathbf{u}/\gamma$. *Relativistic grid*: a one-dimensional, one-momentum-dimension Vlasov–Ampère solver for both species on a fixed (x, u) grid ($N_x = 32$ Fourier modes with the 2/3 mask, 4096 momentum nodes, time step $0.02\,\omega_{pe}^{-1}$), with a prescribed uniform drive and deterministic or shot-noise-like density seeds. It has no Hermite truncation and no closure. *Relativistic Hermite*: a relativistic option of the Hermite–Fourier kernel of the parent solver (momentum-space Hermite expansion in $u = p/mc$; the velocity operator is diagonalized once in the Hermite basis so that streaming and the current remain a matrix function of the same eigendecomposition), combined with the pump frame and the field-scaled closure of Sec. III. The closure rate enters only as a diagonal factor outside the kernel, so the exact Lawson factor is unchanged; streaming stays explicit in the relativistic case, and the Dougherty collision operator and the exact streaming factor are refused (not relativistic). The nonrelativistic limit is bitwise unchanged. *Relativistic PIC*: the independent Dark-JAX-in-Cell code [6] with a relativistic pusher of the Boris/Vay family, the HHS box (1000 cells of $0.04\,c/\omega_{pe}$, $L = 40\,c/\omega_{pe}$), $m_i/m_e = 1836$, $T_i = T_e = 10^{-3} m_e c^2$, quadratic shape functions, $\Delta t = 0.02\,\omega_{pe}^{-1}$; the same code with the Newtonian momentum is the Newtonian reference.

III. METHOD

A. Hermite–Fourier discretization

The velocity dependence is expanded in Cartesian asymmetrically weighted (AW) Hermite functions with independent orders (N_n, N_m, N_p), widths α_s and centres u_s; space is Fourier, with the strict 2/3 dealiasing mask of the parent (which retains $3|m| < N_x$, so every grid size is alias-free for quadratic products). A single call to the SPECTRAX Hermite–Fourier kernel per right-hand-side evaluation receives the physical force fields $\mathbf{E} + \eta\mathbf{E}_D + \mathbf{E}_{\text{drive}}$ and $\mathbf{B} + \eta\mathbf{B}_D$, and the current is evaluated once for both Ampère laws. Time integration is Dopri8 with a PID controller (Diffrax). The kinetic-energy ledger closes only when every forced direction has Hermite order at least 3. The parent is SPECTRAX [7]; the studies reported here were run at the parent commits recorded in each record, all on an unmerged integration branch of open parent changes, not a release (see Data and code availability).

B. Pump frame and basis remaps

In the pump frame the state carries the centres u_s and widths α_s. Each centre follows the total uniform force, $\dot{u}_s = (q/m)_s(\bar{\mathbf{E}} + u_s \times \bar{\mathbf{B}})$, with the box averages including $\eta\mathbf{E}_D$ and the drive; the parent kernel then uses $\mathbf{E} - \bar{\mathbf{E}}$. This is an exact change of variables: the coherent $k = 0$ quiver sits in $u_s(t)$ rather than in high Hermite orders. Widths and residual drifts change only between integration segments, through the parent's exact lower-triangular remap (every moment below the truncation is preserved); triggers fire on $|U - u|/a > 0.3$ or $|a'/a - 1| > 0.05$, with caps $|\Delta u| \le a'$, $a/a' \le 1.1$, $a' \ge 1.1\sigma$.

C. Closures

Truncation needs a closure. The parent closure is a hypercollision $\propto n(n-1)(n-2)$ that leaves Hermite orders 0–2 unchanged. We use a field-scaled form, implemented in the SPECTRAX integration branch,

$$\nu_s = c\left|\frac{q}{m}\right|_s \frac{\sqrt{2N_n}\,\max|E - \langle E\rangle|}{a_s} \times \frac{n(n-1)(n-2)}{(N_n-1)(N_n-2)(N_n-3)}, \quad (10)$$

with strength c (density, momentum and energy rows untouched; a Lorentz-force-scaled variant exists for magnetized runs and is not needed here). For $c = 64$ the closure rate is the stiff eigenvalue of the system: the Jacobian's spectral radius equals the closure rate ν_e (real and negative) to the printed precision, so the explicit step is bounded by $5.17/\nu_e$. We therefore integrate the diagonal closure damping exactly with a Lawson integrating factor [8] wrapped around Dopri8 (integrating-factor and exponential Runge–Kutta methods are reviewed in Ref. [9], and for kinetic equations in Ref. [10]). Lawson transforms $y' = f(t, y)$ with a matrix A by $z = e^{-tA}y$ and advances z with an explicit Runge–Kutta method, so that the stiff part of the Jacobian, $e^{-tA}(\partial f/\partial y)e^{tA} - A$, is removed from the step-size restriction; A may be reevaluated at each step from the local Jacobian (his Algorithm II) [8]. Here $A = -\mathrm{diag}(r)$ is the closure rate frozen at the step start, so with $w = e^{r(t-t_0)}y$ the split is an identity within the step; streaming, the force, the fields and the variation of ν over the step stay explicit. Lawson shows that the construction is A-stable for the linear test problem if the exponential is approximated stably, is stable for $y' = By + u(t, y)$ when the Lipschitz constant of u is small compared with the spectral radius of B, and keeps the order of the underlying Runge–Kutta method when $\exp(hA)$ is approximated to at least that order [8]; here the diagonal exponential is exact. These statements hold for $h \to 0$ at fixed A. For semilinear problems with large A, Hochbruck and Ostermann show that the stiff order of such methods can be lower than the classical one (the Lawson–Runge scheme is only first-order accurate under their stiff-order theory, Example 2.32 and Theorem 2.17 of Ref. [9]), so we do not claim that the order of Dopri8 is retained when $rh \gg 1$; we rely on the measured agreement with the unfactored integration instead. For $c = 64$ the factor reduces the step count by a factor 5.0

to $t = 500$ with ΔK_e and $W_{\rm ext}$ agreeing with the Dopri8 record to 5×10^{-10}; for $c = 4$, where the closure is not stiff, it gives 0.96 (no gain). An exact Hermite-streaming factor was also built and rejected: it gave no step reduction and cost 2.7 times more per step. Table II lists these and other negative results.

D. Collisions

Collisions use the Lenard–Bernstein–Dougherty (LBO) operator [11], $C[f] = \nu\,\partial_{\mathbf{v}}\cdot\big[(\mathbf{v}-\mathbf{u})f + (T/m)\,\partial_{\mathbf{v}} f\big]$, with a velocity-independent frequency ν, an isotropic, velocity-independent diffusion coefficient, and $\mathbf{u}$ and T the flow and temperature moments of f itself (his Eqs. 7–11 and 14). Dougherty shows that it conserves number, momentum and energy and that its only equilibrium is the Maxwellian, and he states its limits: it is a single-species model, accurate for most particles and wrong for the high-energy tail, and the multispecies case is "similar in principle but more cumbersome". We use the conservative Hermite discretization of Hakim et al. [12] with the multispecies forms of Francisquez et al. [13], in two cross-species conventions: LBO-EM (the parent default, the form of Francisquez et al. Eqs. 2.18–2.20: momentum rate equal to the Morse rate and temperature-difference rate equal to twice the Morse rate) and LBO-G with $\beta = 0$ (both rates equal to the Morse rates). The collision frequencies are velocity independent, $\nu_{ee} = \nu_{ei}$, $\nu_{ii} = \nu_{ei}\sqrt{m_e/m_i}$, $m_e\nu_{ei} = m_i\nu_{ie}$; the constancy of ν follows Dougherty's model and is a limitation for the cosmological case (Sec. VI). Collisions do no net work and conserve mass, momentum and energy in the homogeneous controls to round-off.

IV. VERIFICATION AND LIMITS

A. Verification

Table I and Fig. 1 summarize the checks. The linear Landau roots of the Maxwell–Proca–Vlasov system (with $k\lambda_{De} = 0.2$–0.7, $\eta = 0.3$, $\Omega_D = \omega_{pe}$) agree with SciPy Faddeeva roots of $D_L = (Q - \Omega_D^2)(1+\chi) + \eta^2 Q\chi$ to 1.3×10^{-4} across 32 runs, but at $k\lambda_{De} = 0.2$ the damping rate ($\sim 5 \times 10^{-5}$) is not resolved inside the Hermite-limited fit window; only the frequency is validated there. The oscillating two-stream instability (OTSI) that sets the high-k cascade is recovered from the Hermite Floquet analysis against the kinetic dipole-pump dispersion relation of Silin [14] (his Sec. 4, Eqs. 4.4–4.8). For a uniform pump with electron excursion r the perturbation splits into sidebands $\omega + l\omega_0$ coupled by Bessel factors $J_l(kr)$, with Maxwellian susceptibilities χ_s for each species. We evaluate it as the determinant of the sideband matrix $\mathrm{diag}(1+\chi_i) + B\,\mathrm{diag}(\chi_e)B^T$, $B_{lj} = J_{l-j}(kr)$, $|l| \le 4$, with unpumped ions, and compare its purely growing root with the Floquet exponent at 22 unstable points: the worst difference is 1.3 %. In Silin's analysis (cold-plasma Eqs. 3.24–3.35 and the kinetic generalization of Sec. 4) the growth is purely exponential for a pump below the Langmuir frequency, $\omega_0^2 < \omega_{Le}^2 + \omega_{Li}^2$, and oscillatory above it; the rate has a weak-field scaling $(m_e/m_i)^{1/3}\omega_{pe}(kr)^{2/3}$ and a maximum of order $(m_e/m_i)^{1/3}\omega_{pe}$ for $kr \sim 1$ near a harmonic resonance. Nishikawa's coupled-mode theory [15, 16] gives the same classification (non-oscillatory growth for $\omega_0 < \omega_2$, oscillatory for $\omega_0 > \omega_2$, with ω_2 the frequency of the pumped Langmuir wave) and the cube-root dependence on pump power, $y_m = (K/2\omega_2)^{1/3}$ (I, Eq. 40a), but for a fluid plasma with phenomenological damping and a pump energy small compared with the thermal energy (II, Sec. 2). We therefore use it for the classification and the scaling, not as a quantitative reference. With fixed ions every rate vanishes, so the instability needs mobile ions. The relation is known physics; what is new here is the regime (resonant pump with a secularly growing excursion and $T_i = T_e$). Further checks, all in the repository: cold coupled oscillators to 10^{-11}, Proca vacuum and Yukawa limits, the η-sign symmetry, the exact mean pump, a Weibel and filamentation set against an independent kinetic dispersion relation including Proca coupling (within 10^{-3}), magnetized rotation and cold electromagnetic eigenmodes (within 10^{-3}), and Dougherty collisional relaxation against the exact rates (to 10^{-13}). Independent discretizations: a NumPy semi-Lagrangian grid solver (and its Vlasov–Ampere–Proca extension) agrees with the Hermite echo amplitude to 2×10^{-5}; the Dark-JAX-in-Cell PIC code [6] approaches the Hermite Landau fit as cells and particles increase, with a remaining difference of order 1 % in damping set by PIC noise (Fig. 1c); this is not a claim of PIC convergence. GPU and CPU agree to 9.7×10^{-9} on one case only. Gradients of converged dark observables are verified (Table I); reverse mode through Diffrax's `RecursiveCheckpointAdjoint` was found wrong (2–15 %) for many constant-step saves, and `DirectAdjoint` is used.

B. The Hermite truncation instability and the certified windows

The method has one limit that matters for the rest of the paper. In a non-uniform field the truncated AW Galerkin operator $-v\partial_x + (q/m)E\partial_v$ is not anti-self-adjoint; its frozen-field spectrum has growing modes in the top third of the Hermite ladder, with a rate that rises with N_n (Fig. 2a; at $K = 5$, $N_n = 128$, $E_1 = 10^{-2}$ the largest growth rate is 1.6, against $\le 8 \times 10^{-15}$ for the symmetric basis, which is neutral to round-off). This is a numerical instability of the basis, not physics. It is distinct from the physical OTSI, which the Hermite model reproduces and which amplifies finite-k fields strongly at resonance (the Hermite model understates the growth: Nn 32/64/128 agree with each other and end 1.3–4 times below the exact solution). The AW instability is excited

TABLE I. Verification summary. Record files are listed under Data and code availability.

Check	Result (record)
Linear Landau roots, ordinary and dark, $k\lambda_{De} = 0.2$–0.7	worst relative error 1.3×10^{-4} vs. SciPy Faddeeva roots; ledger defect $\leq 3 \times 10^{-18}$
OTSI growth vs. Silin kinetic relation (sideband determinant)	22 unstable points, worst difference 1.3%
Dark-JAX-in-Cell PIC, Landau $k\lambda_{De} = 0.5$	rms $\Delta \ln \lvert E_k \rvert$ falls 0.22→0.13 from 32 to 128 cells; PIC noise limits agreement to ∼1%
GPU vs. CPU, one case ($T = 100$)	max relative difference 9.7×10^{-9} (adaptive trajectories)
AD vs. finite differences	reverse = forward to 2×10^{-13}, analytic to 5×10^{-10} (19 derivatives)

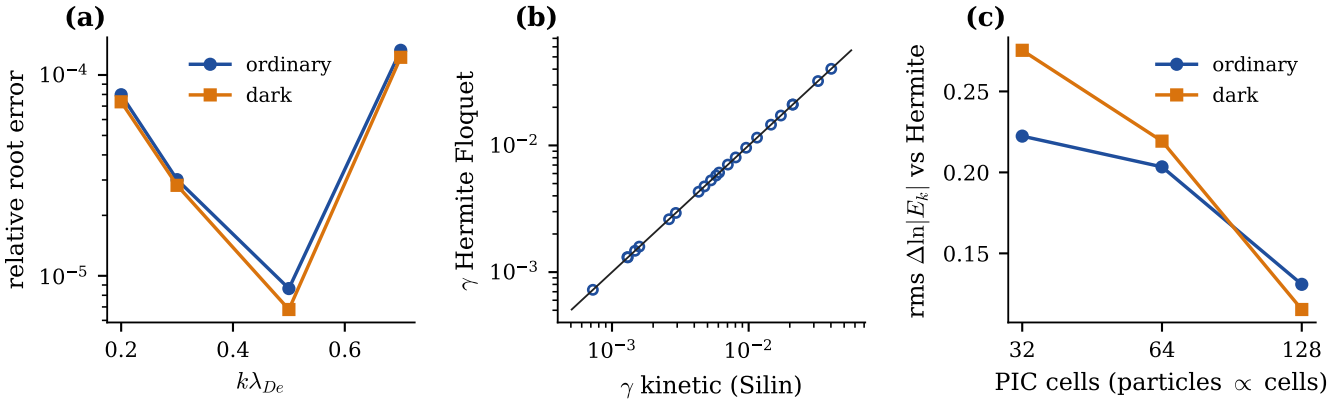


FIG. 1. Verification. (a) Relative error of the fitted complex frequency against the independent kinetic root, ordinary and dark ($\eta = 0.3$), $N_n = 128$, $N_x = 8$. (b) Hermite Floquet growth rate against the kinetic dipole-pump relation of Silin (sideband determinant, see text). (c) RMS difference of $\ln |E_k|$ between Dark-JAX-in-Cell and Hermite on $2 \leq \omega_{pe} t \leq 12$ for Landau damping, for increasing cells and particles.

by the large-amplitude pump and ends the simulation by loss of positivity or by the step budget.

A *certified window* is defined by checks declared before the runs: refinement pairs (N_x, N_n, closure strength) must agree to 10 % in ΔK_e and W_{ext}, the electron distribution must stay positive ($K_s > 0$ and $k = 0\ T_x > 0$), the solver must reach the time, and the energy ledger must close. The window is the minimum of these times. It is therefore a lower bound on where the Hermite results can be trusted, not a statement about physics.

Alternatives were tested and rejected, all under predeclared gates (Table II). Higher resolution on GPU ($N_x \in \{64, 128\}$, $N_n \in \{128, 256\}$, $v_q/v_{te} = 0.1$ and 0.05) failed the declared window-extension check and ended every run inside the instability at 328–595 ($v_q/v_{te} = 0.1$) and 440–751.5 ($v_q/v_{te} = 0.05$): the pairs agree to the end of the shorter run, so this is a failure of the resolution route rather than a disagreement. In a frozen field no closure strength up to 1024 brings the AW growth below 10^{-3} at peak field 4×10^{-3} (39/39 points fail), leaving a floor that grows as $K^{0.15} N_n^{0.30} E^{1.26}$, so a resolution-consistent closure strength does not exist for this closure family. A stable Galerkin form of Dai [17] is unstable in the AW coordinates in float64 at $N_n \geq 64$ (the AW-to-orthonormal change of basis has condition $\sim 2^N$) but stable and accurate against the kinetic OTSI relation in symmetric Hermite functions of width $a/\sqrt{2}$; the symmetric-weight basis itself, and the anti-symmetric square-root form of Issan et al. [18, 19], are linearly neutral to round-off and accurate against the same relation, but were rejected under their declared nonlinear $v_q/v_{te} = 0.1$ gates (symmetric weight: every run blows up at $t = 637$–738 and the refinement pair agrees only to 634; square-root form: departure from the grid reference at 391 and 498, earlier than the symmetric weight). Production therefore stays with the AW basis and the field-scaled closure, and the windows below are its windows.

V. NONRELATIVISTIC BOUNDS: FINITE RESERVOIR AND FINITE COHERENCE

Sections V and VI report results of the nonrelativistic model. Its coherent quiver speed reaches 0.47–0.87 c at the window ends (Sec. X), so the statements below are statements about the model and not about the relativistic plasma; Sec. VII shows how the picture changes.

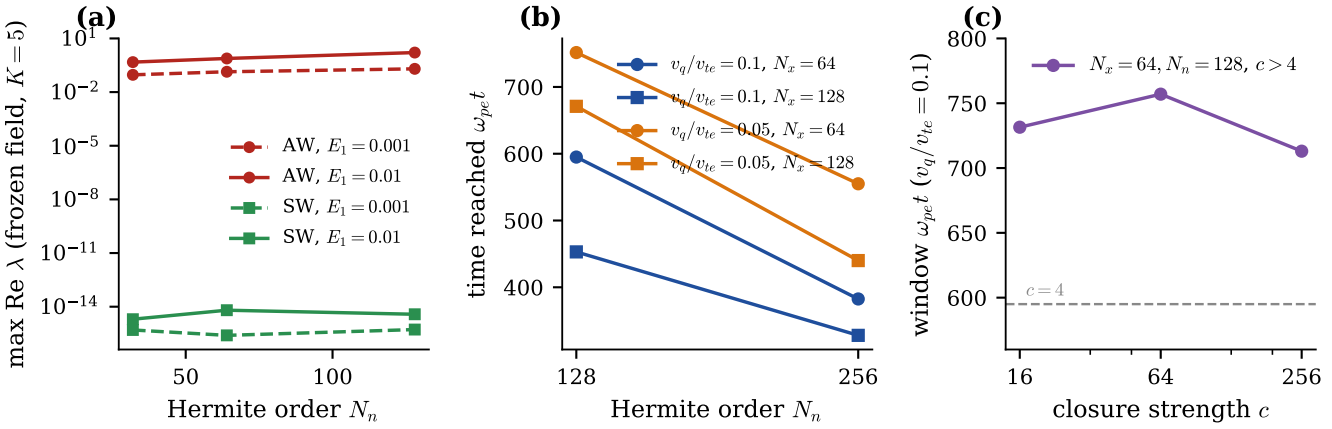


FIG. 2. The limit of the method. (a) Largest real part of the frozen-field spectrum for the AW basis (grows with N_n) and the symmetric-weight basis (round-off), $K = 5$. (b) Time reached by AW runs with closure $c = 4$ at $v_q/v_{te} = 0.1$ and 0.05 for $N_x = 64, 128$ and $N_n = 128, 256$, all ended at the step budget inside the instability: more resolution ends the run sooner. (c) At $v_q/v_{te} = 0.1$, $N_x = 64$, $N_n = 128$ a stronger fixed closure extends the window from 595 to a ceiling of 713–757.

A. Design

Two idealizations of the prescribed-drive calculation are removed. First, in the self-consistent mode the dark photon is a free $k = 0$ Proca mode with $\Omega_D = \omega_L$ whose energy reservoir U_{D0} is finite and is depleted by the transfer; we use $u = U_{D0}/(nT_eV) = 0.1, 1, 10$ with η chosen so that the initial effective force matches the prescribed drive. Second, the dark field has a finite coherence time: the prescribed drive is $E_0 \cos(\omega_L t + \Phi(t))$ with Φ a phase random walk of independent Gaussian frequency samples, giving a Lorentzian line of half-width $1/\tau_c$ and no phase resets. The comparison is always with the exact linear law for the same drive (and, for the reservoir, the linear coupled oscillator $(\bar{E}, \bar{J}, E_D, A_D)$, whose matrix exponential agrees with an ODE solver to 6.6×10^{-11}).

B. Windows at each drive

Table III gives the windows: 1000, 892 and 791.5 for $v_q/v_{te} = 0.03$, 0.05 and 0.07. The windows are set by positivity loss of the $N_x = 64$ members, that is, by the AW truncation instability reaching the resolved range, not by disagreement of the refinement pairs. The strongest certified drive is $v_q/v_{te} = 0.07$ with window 791.5. Drives above that, including $v_q/v_{te} = 0.1$, have no window beyond about 760 (Sec. IV).

C. Linear transfer and the seed back-reaction

At $v_q/v_{te} = 0.07$ the work $W_{\rm ext}$ follows the exact linear resonant law over the whole window (Fig. 3a): the maximum of $|W_{\rm ext}/W_{\rm lin} - 1|$ over four resolutions is 0.68 %, the four members agree on it to 2×10^{-4}, and at the window end $W_{\rm ext} = 381.3$ against $W_{\rm lin} = 384.0$ (in nT_eV). The mean-field frequency shift, measured as the slope of the lock-in phase difference, is at most $6.5 \times 10^{-6}\ \omega_{pe}$ (the declared resolution threshold was $10^{-3}\omega_L$; not resolved). No saturation and no detuning are resolved in this nonrelativistic model. The departure rule (a 10 % departure in the coarsest member confirmed at 5 % in all refinements) is not met.

The residual deficit is not a drive effect. At the three drives it is the same to three digits (it grows as $-10^{-8}\, t^2$) until the OTSI stage, ending at -1.07, -0.88 and -0.68 % at the window ends. A seed-scale test at $v_q/v_{te} = 0.07$ (the initial density fluctuations multiplied by s) shows the deficit scales as s^2 (Fig. 3b): the deficit at $\omega_{pe}t = 400$ is -1.66×10^{-3} for $s = 1$, -1.66×10^{-5} for $s = 0.1$ and 2.3×10^{-9} unseeded (ratio of $s = 0.1$ to $s = 1$: 0.0100). The deficit is therefore the back-reaction of the quiver-advected seed modulation on the mean field (the $\langle \delta n\, \delta E \rangle$ term of the mean-field identity), set by the imposed fluctuation level, not by the pump. A physical fluctuation level different from 5×10^{-4} would rescale it. Our earlier hypothesis that it is a truncation effect of the seedless homogeneous system was tested and rejected: the unseeded deficit is only 1.4×10^{-6} of the seeded one. Only the seed scales 0, 0.1 and 1 at $t \le 400$ were run.

D. Energetics and the onset of the OTSI

Why the transfer stays linear is energetic. At the window end of $v_q/v_{te} = 0.07$, 371 of the $W_{\rm ext} = 381.3\, nT_eV$ resides in coherent $k = 0$ electron quiver kinetic energy; the ions receive 0.05 % of $W_{\rm ext}$. The energy of the $k \neq 0$ electron motion is small and oscillating: with the mean flow removed it sloshes between wave kinetic and field energy with a period of about $67\, \omega_{pe}^{-1}$ near the onset, so a

TABLE II. Negative and amended results that bound the claims (nonrelativistic studies first, then relativistic and PIC). "Declared" means the check was committed before the run and not amended.

Study	Check	Outcome	Consequence
GPU resolution extension	Longer window at higher resolution, $v_q/v_{te} = 0.1$, 0.05	Failed; every run ends at the step budget inside the instability	No new window; a stable basis is needed
Closure-strength route	AW growth below 10^{-3} in a frozen field for some closure strength	Failed; a floor remains	Scaled-closure option not built
Stiffness vs. instability	Which eigenvalue limits the step	Stiff eigenvalue is the closure; the instability (max Re $\lambda = 2$–3) is not	Integrator is not the limit
Lawson factor (closure)	Step reduction at $c = 4$	Failed as declared (no gain)	Useful for $c = 64$ only
Lawson factor (streaming)	Step reduction with an exact streaming factor	Failed	Rejected
Dai / symmetric weight / square root	Declared nonlinear gates at $v_q/v_{te} = 0.1$	Failed (Sec. IV)	AW basis with closure stays
Finite coherence, first version	Declared phase variance	Wrong: realized τ_c was 2× nominal	Corrected version declared and used
Seed deficit hypothesis	"Homogeneous truncation effect"	Rejected	Deficit is seed back-reaction
Collisional steady-state check	Window longer than the record	Declaration defect; declared follow-up used, deviation from $P_{\rm ss}$ 3×10^{-5}	Follow-up result used
Collisions, 5 % agreement	Kinetic equals linear within 5 %	30 of 36 pass; six detuned cases at $\nu_{ei} = 10^{-3}$ fail	Seed effect; seed-scale follow-up passed
Earlier collision pilot	Restart-chunk length; ion heating	Failed as declared (T_i responds to the halved exchange)	Recorded, not amended
Collision-form controls	Several declared checks	Mis-specified or at round-off level	Recorded; ledger closes to 10^{-17}
Relativistic grid, Landau roots	Fitted roots at $k\lambda_{De} = 0.3$, 0.4 against the kinetic roots	First check failed (damping not recovered); a revised check passed (8.9×10^{-5}, 4.5×10^{-4})	Reported as such
Relativistic grid, ledger (D, D2)	Energy defect $\leq 10^{-4}$; ratio 3–5 under Δt halving	Failed: 15 of 16 runs above 10^{-4} (2.3×10^{-4} at $\Delta t = 0.02$, 2.9×10^{-5} at 0.01); ratio 8.0	Grid results carry this defect; pairs agree
Relativistic Hermite, spectrum (c)	Discrete spectrum against the kinetic roots	Failed: discrete spectrum undamped, the nonrelativistic control fails equally (declaration wrong)	Response-function check c2 used (3×10^{-5}, passed)
Relativistic Hermite, 1D2V nonlinear (f4)	Ledger of a nonlinear run with transverse field	Failed: NaN at $\omega_{pe}t \simeq 8$ (truncation instability)	Hypercollision variant f4b (declared after) passed, 6.8×10^{-6}
Relativistic Hermite, horizon (F1)	Certified to $\omega_{pe}t = 2000$	Failed: positivity lost at 1528 and 1584.5	Certified to $\simeq 1530$ only
Relativistic Hermite, noise (F2)	Shot-noise seeds certified to ≥ 400	Failed: $f_e < -10^{-3}$ at 0.5 (103 ppc) and 265–366.5 (2×10^5 ppc)	Hermite does not reach the PIC regime
First PIC comparison (relpic)	ppc and cell refinement stable; heating vs HHS	Failed: +19 % at each of ppc 103 → 412 and Δx 0.04 → 0.02; reading "$\sim 25\,E_{\rm th}$" was a misreading of HHS	Superseded by the convergence study
PIC convergence, C5/C6	Extrapolation and agreement with HHS within 10 %	C6 failed: plateau 0.50 of HHS, Newtonian 1.25; C5 $30.1 \pm 3.8\,E_{\rm th}$, likely an overcount	Factor 2 unresolved; plateau reading used

single time sample is not a heating measure; the average over $200\,\omega_{pe}^{-1}$ (the HHS filter, Sec. VIII) is 5 initial thermal energies at the end of the window. The fluctuating-field amplitude ($\sum_{k\geq 1} |E_k|^2$) reaches about $2077\,E_0^2$ only when the OTSI turns on, at $\omega_{pe}t = 783.5$ ($N_x = 32$) to 862 (higher N_x and N_n members; onset defined in the gates, compared within equal N_x). After that the run ends by positivity loss. Within the window the instability has not extracted a resolvable fraction of the work. *This is the main limitation of the bound*: it says that

TABLE III. Certified windows for the strong-drive set-up with closure $c = 64$ integrated exactly. Members: $(N_x, N_n) = (32, 64), (64, 64), (32, 128), (64, 128)$ and $(32, 64)$ at $c = 32$. The window is the minimum of refinement-pair times, positivity times, times reached and the cap $T = 1000$; the grid column is the masked spectral-grid reference (reported, not part of the window).

v_q/v_{te}	window $\omega_{pe}t$	limiter	max ledger (B)	grid agrees with B until
0.03	1000	cap $T = 1000$	6.6×10^{-12}	1000 / 1000
0.05	892	positivity, $N_x = 64$	3.3×10^{-12}	979.5 / 962.5
0.07	791.5	positivity, $N_x = 64$	9.7×10^{-13}	870 / 863

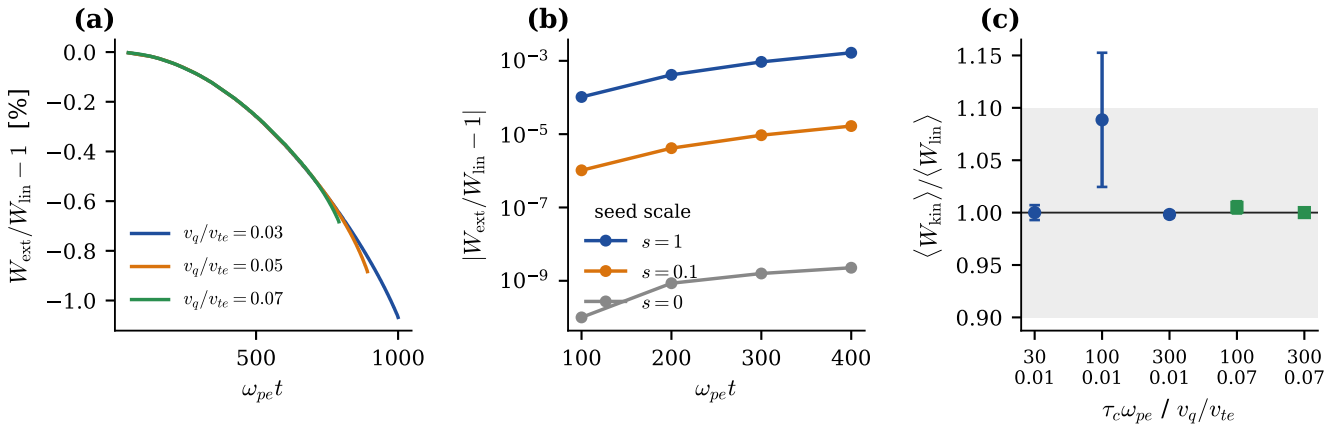


FIG. 3. Finite reservoir and coherence. (a) $W_{ext}/W_{lin} - 1$ for the $N_x = 32$, $N_n = 64$ member at the three drives, each to its certified window; the three curves coincide until the OTSI stage. (b) The deficit at $v_q/v_{te} = 0.07$ for seed scales $s = 1$, 0.1, 0. (c) Finite coherence: ensemble ratio $\langle W_{kin}\rangle/\langle W_{lin}\rangle$ with standard error over six realizations for $\tau_c\omega_{pe} = 30, 100, 300$ at $v_q/v_{te} = 0.01$ (window 1000) and 100, 300 at $v_q/v_{te} = 0.07$ (window 791.5); band: $\pm 10\,\%$.

no saturation appears *before* the cascade of fluctuations starts, and nothing about what happens after.

E. Finite reservoir

In the self-consistent mode the dark energy $U_D(t)$ follows the linear coupled-oscillator prediction at all three drives: within $0.039\,U_{D0}$ (maximum over six cases, $u = 0.1, 1, 10$) at $v_q/v_{te} = 0.01$ and 0.03 over the windows 1000 and 940, and within 0.23 % and 0.81 % of U_{D0} at $v_q/v_{te} = 0.07$ for $u = 1$ and 10 over the window 791.5. The beat times equal π/η ($\eta = 0.049$ and 0.016 for $u = 1, 10$). The accumulated work W_D follows the matched prescribed drive until the split time $t_{split} = 50.5$ and 72 (equal to the linear values): the reservoir gives up 47 % of its energy for $u = 1$ and exchanges energy through beats for $u = 10$, with W_D within 1.0 % and 2.7 % of the linear prediction while it is above 20 % of its running maximum. Finite reservoir depletion is therefore captured by the linear theory; no nonlinear reservoir response is resolved. The ledger closes to 1.7×10^{-9} over all 33 runs of the strong-drive study (declared bound 10^{-6}).

F. Finite coherence

Figure 3c shows the ensemble ratio of the kinetic to the linear same-drive work. At $v_q/v_{te} = 0.01$ (window 1000, with τ_c the field coherence time) the ratios are 1.000 ± 0.007, 1.089 ± 0.064 and 0.998 ± 0.003 for $\tau_c\omega_{pe} = 30, 100, 300$; the value at $\tau_c\omega_{pe} = 100$ is 1.4 standard errors above unity and is consistent with the sampling spread of six realizations. At $v_q/v_{te} = 0.07$ the ratios are 1.0050 ± 0.0058 and 1.0000 ± 0.0038 for $\tau_c\omega_{pe} = 100$, 300, with bounds on the ensemble nonlinear correction (including twice the standard error) of 1.7 % and 0.8 %; individual realizations deviate more (up to 4.7 % and 1.5 % of the peak linear work), with ratios on both sides of unity. The late work rate saturates at about $E_0^2\tau_c\omega_L^2/4$ instead of growing, as in the linear theory. The first coherence runs realized a phase variance dt/τ_c rather than the declared $2\,dt/\tau_c$, i.e. a coherence time twice the nominal; the records were kept and relabelled and a corrected version was declared and run (Table II).

G. Scope of the reservoir and coherence bounds

The bounds hold for one box ($L = 40$), one ion mass, $T_i = T_e$, drives up to $v_q/v_{te} = 0.07$, windows set by the AW instability, fluctuation seeds of one amplitude (5×10^{-4}), a Lorentzian line only, and reservoir and coherence not combined in one run. Not done: a Gaussian line, windows beyond about 790 at 0.07, $v_q/v_{te} > 0.07$ and inhomogeneous dark fields. The grid reference (masked spectral grid) agrees with the Hermite coarse member until 870 and 863 at $v_q/v_{te} = 0.07$ (Table III).

VI. NONRELATIVISTIC COLLISIONAL ABSORPTION

A. Linear drag law

With collisions the uniform relative velocity relaxes at the Morse rate, identical in LBO-EM and LBO-G ($\beta = 0$),

$$\nu_{\rm eff} = \frac{\nu_{ei}}{2}\Big(1 + \frac{m_e}{m_i}\Big), \tag{11}$$

so the uniform driven cold response obeys $\dot J = \omega_L^2(\bar E + E_D) - \nu_{\rm eff} J$ and the steady absorbed power per volume is

$$P_{\rm ss} = \frac{\omega_L^2\,\nu_{\rm eff}\,\omega^2 E_0^2}{2\,[(\omega_L^2 - \omega^2)^2 + \nu_{\rm eff}^2\omega^2]}, \tag{12}$$

$$P_{\rm ss}^{\rm res} = \frac{\omega_L^2 E_0^2}{2\nu_{\rm eff}}, \tag{13}$$

with the transient decaying as $\exp(-\nu_{\rm eff} t/2)$ (1/e time $4/\nu_{ei}$). This is the cold Drude response of Dubovsky and Hernández-Chifflet [2], who write the friction as $-m_e\nu(u_e - u_i)$ with a Spitzer-like ν and find a shielding suppression $\propto m^2/\omega_p^2$ off resonance; and it matches the non-resonant absorption rate of McDermott and Witte [3]. Their ν corresponds to $\nu_{\rm eff}$ here. In our model ν is *constant*: the Spitzer scaling $\nu \propto T_e^{-3/2}$ is absent.

B. Design and gates

The set-up is that of the strong-drive study (mobile ions, pump frame, $c = 64$ with Lawson, $T = 1000$) with Dougherty e–e, e–i and i–i collisions in both conventions. The grid is $v_q/v_{te} = 0.01, 0.03, 0.05$ by $\nu_{ei} = 10^{-3}, 3\times10^{-3}, 10^{-2}, 3\times10^{-2}$ by $\omega = \omega_L, 0.9, 1.1$ (in ω_{pe}): 192 runs in all, including $N_x = 64$ and $N_n = 128$ refinement members and unseeded controls. Checks were declared before the runs. Every LBO-EM case reaches $t = 1000$ with refinement pairs agreeing to the end and a refinement spread of the late rate $\le 2.2 \times 10^{-4}$. The nonrelativistic window ($T_e/T_e(0) < 90$, i.e. $v_{te} < 0.3c$) is shorter only for $v_q/v_{te} = 0.05$ resonant. Rates are evaluated at $t_b = 996.5$.

C. Kinetic against linear

Table IV and Fig. 4 give the result. Where collisions act and the linear transient has decayed, the kinetic absorption equals the linear drag law: at $\nu_{ei} = 3 \times 10^{-2}$ the late rate is within 0.4 % of $P_{\rm ss}$ for all three frequencies and drives, and the steady-state law holds in 12 of 12 cases where $P_{\rm lin}$ is within 10 % of $P_{\rm ss}$. For the other cases the kinetic rate equals the exact *transient-inclusive* linear law to 5 % except in six cases, but the steady state is not reached within $t = 1000$: for $\nu_{ei} = 10^{-3}, 3\times10^{-3}, 10^{-2}$ the resonant rate is only 0.20, 0.48, 0.89 of $P_{\rm ss}$ (the amplitude transient has 1/e time $4/\nu_{ei}$). The sustained law is therefore tested directly at $\nu_{ei} = 3 \times 10^{-2}$ (and 10^{-2} detuned), and through the linear ODE for the rest. The 5 % agreement check passes in 30 of 36 cases.

The six failures are the detuned $\nu_{ei} = 10^{-3}$ cases, where the kinetic late rate is 8–14 % high (mean +14 % at 0.9 and +9 % at 1.1), independent of v_q/v_{te}. In the unseeded controls the kinetic and linear laws agree to 3×10^{-8}, and a seed-scale follow-up (declared after the main analysis and before its runs) shows the departure scales as s^2 (ratios 0.0105 and 0.0099 for $s = 0.1$, against $s^2 = 0.01$): it is the same seed back-reaction as for the strong-drive study, large only where $W_{\rm ext}$ is itself tiny. The resonant late rate is low by 1.5–4 % at $\nu_{ei} = 10^{-3}$ (0.963 at the lowest) and by ≤ 0.2 % at 3×10^{-2}; at $v_q/v_{te} = 0.05$ resonant the deficit (3.7 %) is larger than at 0.01 and 0.03, but drive amplitude and heating are not separable in these data, so a few-percent heating or drive correction is allowed but not attributed.

D. Deposition law

Fitting $\ln P = a + p\ln(v_q/v_{te}) + q\ln\nu_{ei}$ over the 12 cases per frequency (standard errors from the residual scatter, so they include the transient-state curvature) gives for resonant, 0.9 and 1.1: $p = 1.99 \pm 0.10$, 2.00 ± 0.03, 2.00 ± 0.02 (kinetic), against 2 (linear, exact), and $q = -0.51 \pm 0.05$, 0.75 ± 0.02, 0.88 ± 0.01 (kinetic) against -0.52, 0.79, 0.90 (linear). The amplitude scaling is quadratic. The ν_{ei} scaling follows the transient-inclusive linear law, not a pure power: the detuned rate grows with ν (off-resonant absorption proportional to ν, until the beat transient decays), and the resonant rate falls with ν with an exponent near $-1/2$ because the steady state is not reached. The geometric-mean correction of kinetic to linear power is $\kappa = 0.985 \pm 0.003$, 1.040 ± 0.017, 1.025 ± 0.011. The domain of the law is $v_q/v_{te} \le 0.05$, $\nu_{ei} \in [10^{-3}, 3\times10^{-2}]\omega_{pe}$, $\omega \in \{0.9, \omega_L, 1.1\}\omega_{pe}$, $t \le 1000$, one box and mass ratio, one seed level.

E. Collision form and heating

EM and G ($\beta = 0$) agree in $W_{\rm ext}$ to 2.5×10^{-6} and the late-rate ratio differs by $\le 7 \times 10^{-6}$ (agreement

TABLE IV. Late-time absorbed power, kinetic ($N_x = 32$, $N_n = 64$, LBO-EM) against the exact transient-inclusive linear law, $v_q/v_{te} = 0.03$. $T_e/T_e(0)$ at t_b; last column: kinetic equals linear within 5 %.

ω/ω_{pe}	ν_{ei}/ω_{pe}	$P_{\rm lin}/P_{\rm ss}$	$P_{\rm kin}/P_{\rm ss}$	$P_{\rm kin}/P_{\rm lin}$	$T_e/T_e(0)$	5 % check
res.	0.001	0.20	0.192	0.9830	35.2	pass
res.	0.003	0.48	0.472	0.9841	68.6	pass
res.	0.01	0.89	0.877	0.9908	83.6	pass
res.	0.03	1.00	0.996	0.9979	48.1	pass
0.9	0.001	2.00	2.279	1.1406	1.02	FAIL
0.9	0.003	1.54	1.584	1.0281	1.05	pass
0.9	0.01	1.11	1.106	0.9994	1.12	pass
0.9	0.03	1.00	1.003	1.0000	1.32	pass
1.1	0.001	1.38	1.496	1.0857	1.02	FAIL
1.1	0.003	1.21	1.224	1.0145	1.05	pass
1.1	0.01	1.04	1.037	0.9996	1.14	pass
1.1	0.03	1.00	1.000	1.0000	1.38	pass

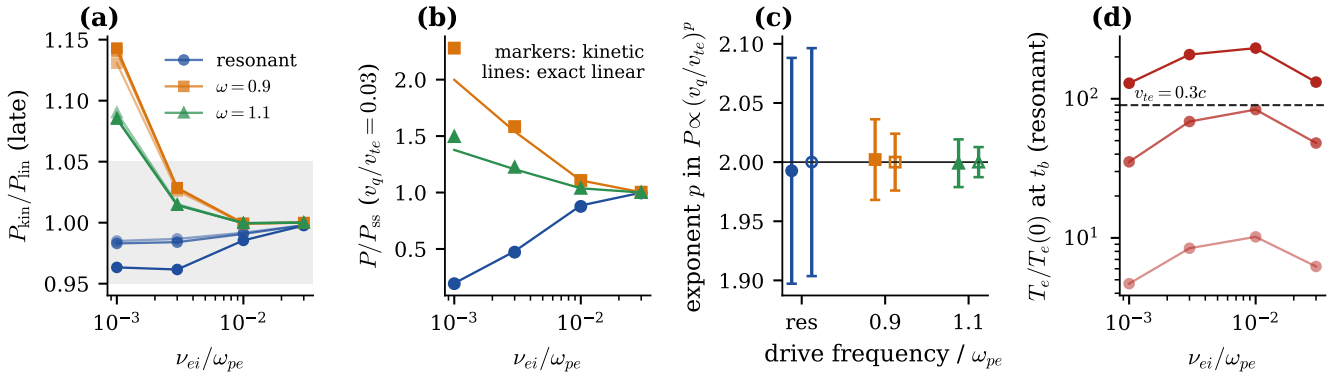


FIG. 4. Collisional absorption. (a) Ratio of the kinetic to the exact linear late-time power for $v_q/v_{te} = 0.01, 0.03, 0.05$ (increasing opacity) and three drive frequencies; band: the 5 % gate. (b) Late power relative to the steady state $P_{\rm ss}$ at $v_q/v_{te} = 0.03$, kinetic (markers) and linear (lines). (c) Fitted exponent p of the drive amplitude, kinetic (filled) and linear (open), with standard errors; the solid line marks $p = 2$. (d) Electron heating at t_b for the resonant cases, same opacity code as (a); dashed: $v_{te} = 0.3c$, beyond which the nonrelativistic model is outside its validity.

checks pass in 36 of 36 cases); electron kinetic energy and temperature differ by $\leq 8 \times 10^{-3}$. Ion temperature differs by up to 31 % and the electron–ion energy exchange is halved, as expected: the absorption is set by the momentum friction, identical in the two forms, while the exchange is the temperature-difference rate. The ion-heating check of the earlier pilot failed as declared for this reason (Table II). Electron heating is large: $T_e/T_e(0)$ reaches 231 (Fig. 4d), beyond $v_{te} = 0.3c$ for $T_e/T_e(0) > 90$, i.e. outside the nonrelativistic model. Heating does not change the rate in this model because ν is constant ($R_{\rm late}$ stays within 1 % of unity while $T_e/T_e(0)$ rises from 7 to 70 at $v_q/v_{te} = 0.03$, $\nu_{ei} = 10^{-2}$, resonant). Ion heating is small ($T_i/T_i(0) \leq 2.8$) and the $e \to i$ exchange is 4×10^{-5} to 1.6×10^{-2} of $W_{\rm ext}$, growing in proportion to ν_{ei}. The steady-state self-check of the linear theory had a declaration defect; its declared follow-up passes (Table II).

F. What is needed for a cosmological statement

Our collisions fix ν at a constant. A cosmological deposition law needs (i) the velocity-dependent Coulomb frequency $\nu(T_e) \propto T_e^{-3/2}$, which couples the heating to the absorption and lowers it; (ii) radiative and Compton cooling; (iii) cosmological expansion and ionization history; (iv) relativistic electrons, since $T_e/T_e(0)$ exceeds the nonrelativistic limit; (v) the $k \neq 0$ coupling of a real dark-photon field; and (vi) a steady-state measurement at $\nu_{ei} \leq 10^{-2}$ resonant, which needs $t \gg 4/\nu_{ei}$. ν_{ee} and ν_{ii} are order-of-magnitude choices that do not enter the uniform drag.

VII. RELATIVISTIC DETUNING AND THE COLLAPSE OF THE MEAN-FIELD RESPONSE

A. Verification of the relativistic models

The relativistic grid reproduces the exact relativistic homogeneous problem and, in the linear regime, the kinetic Landau roots: at $k\lambda_{De} = 0.3$ and 0.4 the fitted frequency differs from the root by at most 8.9×10^{-5} and the damping by 4.5×10^{-4} (relative), closer to the relativistic than to the nonrelativistic root; a first version of this check failed (Table II). Its energy ledger does not close as tightly as that of the Hermite solver: the relative defect is 2.3×10^{-4} at $\Delta t = 0.02\,\omega_{pe}^{-1}$ (declared bound 10^{-4}, failed in 15 of 16 runs) and 2.9×10^{-5} at $\Delta t = 0.01\,\omega_{pe}^{-1}$, with a ratio 8.0 between the two outside the declared range 3–5 (Table II); refinement pairs in N_x, momentum nodes and Δt agree to $\omega_{pe}t = 1000$ (declared 10 % rule).

The relativistic Hermite model reduces to the exact homogeneous relativistic law, including the pump frame, to 2.5×10^{-9} in $\bar{E}$ and 7.8×10^{-10} in $W_{\rm ext}$ for $\omega_{pe}t \leq 500$, a check that discriminates: the nonrelativistic law misses $W_{\rm ext}$ by 38 %. Its right-hand side agrees with the nonrelativistic solver in the limit $c \to \infty$ to 1×10^{-15}, a single particle gyrating in a transverse magnetic field matches the exact relativistic frequency to 2×10^{-10}, the magnetic force does no work to 3×10^{-15}, and the ledger including $(\gamma - 1)mc^2$ closes to 6.6×10^{-10} for a pumped electron at $\simeq 0.13\,c$ and to 8×10^{-9} at $v = 0.83\,c$. In the production runs below the ledger closes to 1.8×10^{-8} for $\omega_{pe}t \leq 400$. The nonlinear 1D2V check with a transverse field failed as declared through the truncation instability of Sec. IV (not a relativistic effect: frozen-field growth is 4.7 relativistic against 4.9 nonrelativistic), and a variant with hypercollision declared afterwards closes to 6.8×10^{-6} (Table II).

B. Collapse of the mean field

The mean-field work obeys, in the homogeneous limit, a driven oscillator whose frequency depends on the electron γ through the plasma frequency; a coherent quiver speed of 0.1–0.2 c shifts it by more than the resonance width $\sim 1/(\omega_{pe}t)$. In the exact homogeneous problem the relativistic $W_{\rm ext}$ is 0.0056 of the nonrelativistic one at $\omega_{pe}t = 1000$. In the spatially resolved relativistic runs the departure of $W_{\rm ext}$ from the exact nonrelativistic linear law by 10 % occurs at $\omega_{pe}t = 392$ for the deterministic grid and 392.5 for the Hermite model, identically in all 8 production runs ($N_x \in \{32, 64\}$, $N_n \in \{64, 128\}$, closure strength 16 and 64; 8 of 8 at exactly this value). With broadband shot-noise-like seeds on the grid, the collapse time is 367–392.5 (the seeds that remain valid long enough); nonrelativistic grids with the same seeds give 769.5–866.5, and the nonrelativistic Hermite model of Sec. V shows none within its window. At the weak drive $v_q/v_{te} = 10^{-3}$ the relativistic grid collapses at 1309.5 (deterministic) and 1413.5 (noise), while the nonrelativistic grid does not collapse to $\omega_{pe}t = 3000$ (mean-field energy still 0.71 of the linear value there, against 0.50 for the relativistic grid). The mean-field energy is 0.85 of the linear value at $\omega_{pe}t = 500$ and 0.041 (grid) and 0.041 (Hermite) at 1000 (Table V, Fig. 5a). HHS state that the resonance is lost at about $\omega_{pe}t = 800$ and show a mean-field energy of $\sim 10^{-3}$ there, $\lesssim 10^{-4}$ at 1000 (units $n_e m_e c^2$). The timing of the relativistic collapse is consistent with this; the absolute late-time level is not: the deterministic continuum retains $3.7\,E_{\rm th}$ (1.9×10^{-3}) at 1000, whereas the PIC retains $0.25\,E_{\rm th}$ (Sec. VIII).

The collapse in the PIC occurs earlier in both pushers ($\omega_{pe}t = 273$–286 relativistic, 292–311.5 Newtonian, $N_{\rm ppc} = 412$; Table V) than in the deterministic continuum models: the PIC contains the noise that the deterministic runs lack, and the noise alone suffices to depart from the linear law by 10 % in work even with the Newtonian pusher. Relativity makes the PIC mean-field collapse faster at later times (mean-field energy over linear at $\omega_{pe}t = 800$: 0.008 relativistic, 0.076 Newtonian), and the mean-field energy at 1000 is 0.003 of the linear value in both. The statement that relativity sets the timing is therefore made in the deterministic models, where relativity is necessary for a collapse before $\omega_{pe}t \simeq 800$ and sufficient to produce one at 392; in the PIC the timing mixes relativity with noise.

VIII. ELECTRON HEATING: RELATIVISTIC PIC AND THE HHS CURVE

A. The HHS measure and its re-digitization

The energy "$\Delta E/n_e m_e$" of the HHS Figs. 2 and 3 is the change of the thermal energy density $n_\alpha m_\alpha v_{th}^2/2$ of each species (SM Eqs. S35 and S36), with v_{th}^2 the one-dimensional (x) velocity variance about the mean; SM footnote 8 states that the plotted temperatures measure the total kinetic energy with the $k = 0$ component removed, so that energy already thermalized cannot be told apart from energy in $k \neq 0$ oscillations. The coherent $k = 0$ quiver energy is therefore not counted; the mean field enters separately as $\langle E^2 \rangle/2$. All curves are filtered with a moving average over $200\,\omega_{pe}^{-1}$. Our records carry the box-mean-flow-subtracted one-dimensional electron variance and the mean field, and every curve of Fig. 5 uses the same trailing $200\,\omega_{pe}^{-1}$ average and the same units, $E_{\rm th} = 5 \times 10^{-4}\,n_e m_e c^2$. HHS use the Vay pusher, 1000 cells of $0.04\,c/\omega_{pe}$ and 2×10^2 to 2×10^5 particles per cell, chosen so that shot noise is comparable to the driven energy at $\omega_{pe}t \approx 2.4$ (SM Sec. II, Eq. S38).

We re-digitized their Fig. 2 (upper panel) from the vector figure rendered at 600 dpi, calibrating the axes on the detected grid lines; the dotted initial thermal energy reads 5.02×10^{-4} against the nominal 5×10^{-4}. The black solid line, the $k \neq 0$ electron energy change, reads

TABLE V. Resonant drive $v_q/v_{te} = 0.03$: collapse time of $W_{\rm ext}$ (10 % departure from the nonrelativistic linear law), mean-field energy over its linear value, and $k \neq 0$ electron heating (trailing $200\,\omega_{pe}^{-1}$ average, units of $E_{\rm th} = 5 \times 10^{-4}\, n_e m_e c^2$) at $\omega_{pe}t = 1000$, with the time to which the model is valid (positivity $f_e > -10^{-3}$, ledger and refinement checks). †Shot-noise seeds: the numbers are at 1000 but the runs lose positivity earlier, so they are not results; ranges are over particle numbers 2×10^4–2×10^5 per cell and $N_x = 32$–128 (grid). PIC: seed mean at $N_{\rm ppc} = 412$ (5 seeds rel., 2 Newtonian); the PIC collapse is the range over seeds. HHS: values read from their Fig. 2 and SM.

model	collapse time	$\bar{E}^2/2$ over linear, $t = 1000$	$\Delta E_e/E_{\rm th}$, $t = 1000$	valid to
Hermite, nonrel. ($N_x = 32, N_n = 64$)	none (≤ 1000)	0.96	2.5	1000
grid, nonrel., shot noise†	769.5–866.5	0.12–0.26	99–209	240–394
grid, rel., deterministic	392	0.041	0.056	1000
Hermite, rel., deterministic	392.5	0.041	0.056	1528–1584.5
grid, rel., shot noise†	367–392.5	0.008–0.30	15–51	240.5–407.5
PIC, rel. (ppc 412)	273–286	0.003	24.6	—
PIC, Newtonian (ppc 412)	292–311.5	0.003	62.7	—
HHS (digitized)	loss of resonance ~ 800	—	50.1	—

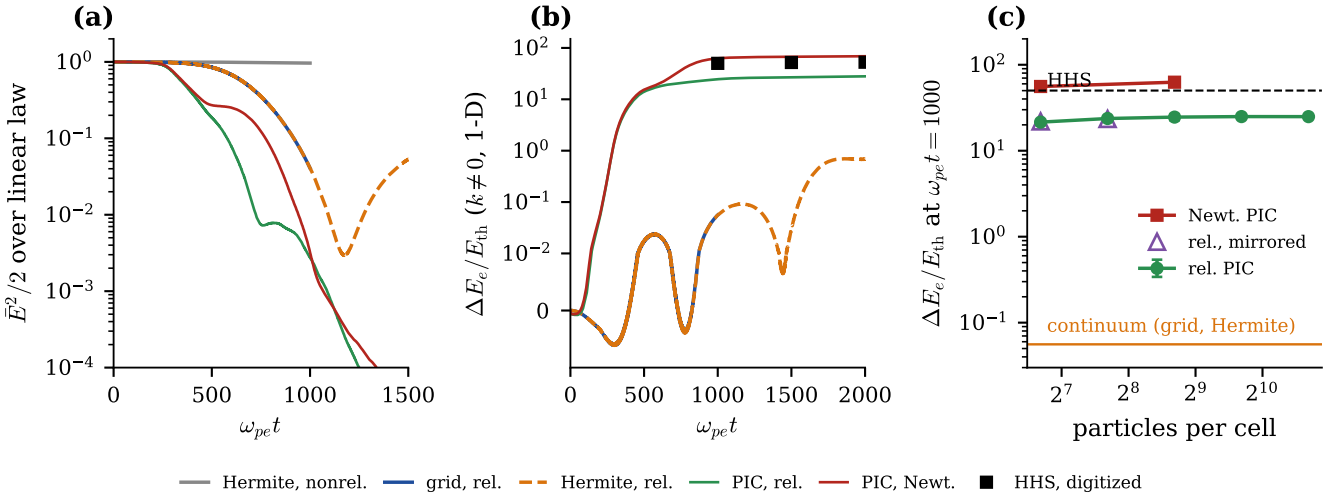


FIG. 5. Resonant drive $v_q/v_{te} = 0.03$. (a) Mean-field energy $\bar{E}^2/2$ over the exact linear law (trailing $200\,\omega_{pe}^{-1}$ average): nonrelativistic Hermite (to its window), relativistic grid and relativistic Hermite (certified), relativistic and Newtonian PIC at 412 particles per cell (seed means). (b) HHS measure: change of the $k \neq 0$ one-dimensional electron energy over $E_{\rm th}$ (symmetric logarithmic scale); squares: HHS Fig. 2, re-digitized. The continuum curves coincide until 1000 and stay below $\sim 1\,E_{\rm th}$ to 2000. (c) Heating at $\omega_{pe}t = 1000$ against particles per cell (error bars: seed standard deviation), the relativistic PIC plateau, the Newtonian PIC, mirrored loading, the HHS value and the continuum value.

$50.1\,E_{\rm th}$ at $\omega_{pe}t = 1000$ and $52.7\,E_{\rm th}$ at 2000 (line-width uncertainty about 3 %). An earlier reading of $\sim 25\,E_{\rm th}$ in our work was a misreading and is superseded (Table II).

B. Convergence of the relativistic PIC

Table VI and Fig. 5c give the heating of the relativistic PIC at $\omega_{pe}t = 1000$ over the ladder of particles per cell, cell size, shape function and time step, against the Newtonian pusher. The heating rises from $21.5\,E_{\rm th}$ at 103 particles per cell to $24.6\,E_{\rm th}$ at 412 and then stays flat: the successive changes from 412 to 824 and from 824 to 1648 particles per cell are 1.4 % and 0.1 %, and halving the cell to 0.02 and $0.01\,c/\omega_{pe}$ changes it by 0.8 % (24.1 and $24.3\,E_{\rm th}$), quintic shape functions give $25.7\,E_{\rm th}$ at 412 particles per cell and halving Δt gives 21.1 at 103 (against 21.5). The plateau is $\simeq 25\,E_{\rm th}$, i.e. 0.50 of the re-digitized HHS value. The declared convergence gates passed (successive differences within 5 % or twice the seed spread; the tolerances include the seed allowance, which at 412 particles per cell is $2.0\,E_{\rm th}$ over five seeds); at $\omega_{pe}t = 2000$ the heating is $28.0 \pm 3.7\,E_{\rm th}$, against 52.7 in HHS. Peak mean-field energy is $11\,E_{\rm th}$.

TABLE VI. PIC electron heating $\Delta E_e/E_{\rm th}$ (trailing $200\,\omega_{pe}^{-1}$ average) for $v_q/v_{te} = 0.03$: relativistic unless stated, quadratic shape, $\Delta x = 0.04\,c/\omega_{pe}$, $\Delta t = 0.02\,\omega_{pe}^{-1}$ except where varied. ±: standard deviation over seeds. HHS: re-digitized.

variation	$\omega_{pe}t = 1000$	2000
103 ppc (3 seeds)	21.5 ± 0.6	—
206 ppc	23.7	—
412 ppc (5 seeds)	24.6 ± 2.0	28.0 ± 3.7
824 ppc	25.0	—
1648 ppc	24.9	—
$\Delta x = 0.02$, 0.01 (103 ppc)	24.1, 24.3	—
quintic shape (412 ppc)	25.7	—
$\Delta t = 0.01$ (103 ppc)	21.1	—
mirrored, 103 ppc (2 seeds)	21.9, 22.6	—
mirrored, 206 ppc	23.4	—
Newtonian, 103 ppc (3 seeds)	55.9	—
Newtonian, 412 ppc (2 seeds)	62.7 ± 9.4	68.9
HHS (digitized)	50.1	52.7

With the Newtonian pusher, the same code at 412 particles per cell gives $62.7 \pm 9.4\,E_{\rm th}$ (two seeds; 55.9 at 103, three seeds) and 68.9 at 2000: 1.25 times the HHS value at 1000, and 2.5 times the relativistic result; the relativistic-to-Newtonian ratio is 0.39 at $\omega_{pe}t = 1000$ and 0.41 at 2000. Relativity therefore reduces the heating by a factor of order two and a half, and the Newtonian pusher cannot be used to infer it.

C. Extrapolation, comparison gate and noise

A declared extrapolation to infinite particle number and zero cell size, multiplying the 412-particle value by fitted ratios (fits in $N_{\rm ppc}^{-1}$ and $N_{\rm ppc}^{-1/2}$ and in Δx^2), gives $30.1 \pm 3.8\,E_{\rm th}$ (C5). We regard it as an overcount: the ppc and cell-size trends are the same $\sim 15\,\%$ rise between 103 and 412 particles per cell and the combined point (412 particles per cell and $\Delta x \le 0.02$) was not run, so the product of the two corrections counts one effect twice; the direct plateau reading is $\simeq 25\,E_{\rm th}$. The declared comparison with HHS, agreement within 10 % including the digitization uncertainty (C6), *failed*: the relativistic value is 0.50 of HHS (0.60 with the extrapolated value), the Newtonian 1.25. Both outcomes are recorded as declared.

A declared follow-up tested whether the heating is limited by long-wavelength shot noise: velocities were loaded in exactly cancelling $(v, -v)$ pairs on adjacent particles, which removes the sampled long-wavelength current noise. The heating is unchanged (21.9 and $22.6\,E_{\rm th}$ at 103 particles per cell against 21.5 ± 0.6; 23.4 at 206 against 23.7) and rises with particle number, with no fall as $1/N_{\rm ppc}$. The heating is therefore a converged plateau, not a noise level that decreases with sampling. The test does not exclude noise at the cell scale, which the pairing does not cancel.

D. What is and is not resolved

The relativistic PIC reproduces the HHS heating to a factor of about two with a rise at $\omega_{pe}t \sim 200$–600 and a flat late-time level like the digitized curve (Fig. 5b), and shows that a Newtonian pusher would overestimate it. We do not know why our plateau is half of the digitized HHS curve and we make no claim that either is wrong. Candidate causes that we have not tested: their fifth-order shape functions and the SHARP scheme against our quadratic (a quintic test at 412 particles per cell changes ours by only 25.7 against 24.6), their particle number per cell and loading, a different pusher (Vay against our Boris/Vay-family implementation), the exact definition of the plotted energy (our $k \neq 0$ one-dimensional electron variance with the box-mean flow removed, by the SM footnote), and the digitization itself (about 3 %).

IX. CONTINUUM METHODS AGAINST THE PIC HEATING: AN OPEN QUESTION

Both relativistic continuum models give a heating that is orders of magnitude below the PIC. The deterministic grid and Hermite runs agree to three digits: $0.018\,E_{\rm th}$ at $\omega_{pe}t = 500$ and $0.056\,E_{\rm th}$ at 1000, identical in all 8 production runs, and the Hermite model stays below $0.71\,E_{\rm th}$ up to $\omega_{pe}t = 2000$ (Fig. 5b); the relativistic PIC plateau is about 445 times the continuum value. The Hermite model is certified (positivity $f_e > -10^{-3}$, ledger $\le 10^{-4}$, refinement pairs within 10 %) to $\omega_{pe}t = 1528$–1584.5 and not to the declared 2000 (Table II). Extending the non-relativistic grid to $k\lambda_{De} = 1.69$ ($N_x = 1024$) raises the deterministic gain only to $2.3\,E_{\rm th}$ at 1000, so the Fourier range alone does not account for the difference.

The remaining candidate is the fluctuation level. When the continuum models are given broadband density noise equivalent to a PIC (grid: 2×10^4–2×10^5 particles per cell; Hermite: the same recipe), they lose positivity ($f_e < -10^{-3}$) at $\omega_{pe}t = 240.5$–407.5 (grid) and 265–366.5 (Hermite at 2×10^5 per cell), and at the PIC particle number at once: the grid with 205 particles per cell at $\omega_{pe}t = 0.5$, and the Hermite model with 103 at $\omega_{pe}t = 0.5$ (positivity stop at $\omega_{pe}t = 160$); the non-relativistic grid with shot noise at $k\lambda_{De} \le 1.69$ loses it at $\omega_{pe}t = 69$–163.5 with $13.5 \times 10^{-3}\,E_{\rm th}$ of heating. The heating values at 1000 of the grid with shot noise (Table V) are therefore not results: they were produced after validity was lost, and one cannot say whether they are physical. The pointed question is therefore: which fluctuations, at what scales and amplitudes, are present in the PIC at $\omega_{pe}t \sim 200$–400 that the deterministic seeds and the positivity-limited noisy seeds of the continuum models do not contain, and can an admissible (positive, converged) continuum initial condition reproduce the heating without PIC noise? The mirrored-loading test excludes long-wavelength current noise as the seed but not

cell-scale noise, and the continuum codes do not represent cell-scale noise at all. A converged continuum calculation with noise would require a positivity-preserving discretization (the Hermite truncation instability of Sec. IV is of the same kind) and is beyond what the models here can certify.

X. DISCUSSION AND CONCLUSIONS

What the results establish. (i) *Relativistic detuning.* In both relativistic continuum models, which agree with each other to the plotted accuracy and are certified past the time of interest, the resonant transfer collapses by 10 % from the linear law at $\omega_{pe}t \simeq 392.5$ for $v_q/v_{te} = 0.03$ and 1309.5–1413.5 for 10^{-3}; the nonrelativistic models give 769.5–866.5 or no collapse. The mean-field energy is 0.041 of its linear value at 1000. This is the timing and the mechanism of the suppression of the mean-field response, and it requires no nonlinear plasma process: it is the relativistic mass increase of the coherently quivering electrons. In the PIC the collapse is earlier and noise-assisted in either pusher, and the late mean-field level of the PIC is lower than that of the continuum models (Sec. VII). (ii) *Heating magnitude.* An independent relativistic PIC converges, within the ladders tested, to $\simeq 25\,E_{\rm th}$ at $\omega_{pe}t = 1000$, about half of the re-digitized HHS value (50.1); the Newtonian pusher gives 62.7 ± 9.4, i.e. 2.5 times the relativistic result and 1.25 times HHS. The factor ~ 2 with respect to HHS is not explained. (iii) *Continuum methods.* The relativistic grid and Hermite models resolve the detuning, and give $0.056\,E_{\rm th}$ of heating, which is orders of magnitude below the PIC; noisy continuum calculations lose positivity before heating develops (Sec. IX). The origin of the PIC heating in terms of fluctuations is open.

Earlier nonrelativistic results and their scope. The nonrelativistic Hermite–Fourier solver was verified (Table I) and its limit is a numerical truncation instability of the asymmetrically weighted basis, which bounds the certified windows to 1000, 892 and 791.5 at $v_q/v_{te} = 0.03$, 0.05, 0.07; no higher resolution, stronger closure or alternative basis we tried removes it (Table II). Inside the windows the transfer follows the linear law to 0.68 % with no resolved detuning (bound $6.5 \times 10^{-6}\,\omega_{pe}$), a finite dark reservoir and a finite coherence time leave the linear prediction unchanged to 0.81 % of the reservoir energy and 1.7 % of the ensemble work, and collisional absorption is the linear Drude law with $\nu_{\rm eff} = (\nu_{ei}/2)(1 + m_e/m_i)$ (within 0.4 % of steady state at $\nu_{ei} = 3 \times 10^{-2}\omega_{pe}$, $p = 1.99 \pm 0.10$). These results are statements about a nonrelativistic model. Its coherent quiver speed, $\sqrt{2W_{\rm ext}/(nm_e)} = 0.47$, 0.70 and $0.87\,c$ at the window ends, is outside the range in which Newtonian dynamics holds, and the relativistic runs show that in this regime the nonrelativistic law fails already at $\omega_{pe}t \simeq 400$ in the resonant case. The nonrelativistic bounds therefore remain valid for the model they are derived in (and for the early part of the evolution, $\omega_{pe}t \lesssim 200$ at $v_q/v_{te} = 0.03$, where the quiver speed is $\lesssim 0.1\,c$), but they do not bound relativistic plasmas. The collisional law is not tested relativistically; the electron temperature in those runs reaches $v_{te} > 0.3c$ (Fig. 4d).

Limits. All models are one-dimensional with a box $L = 40\,c/\omega_{pe}$ and a prescribed uniform drive, $m_i/m_e = 1836$ and $T_i = T_e$; the cosmological law needs a temperature-dependent collision frequency, cooling, expansion and the $k \neq 0$ coupling of a real dark-photon field (Sec. VI). The relativistic Hermite model has no collisions, and the grid ledger fails its declared bound at $\Delta t = 0.02\,\omega_{pe}^{-1}$ (Table II). The PIC results are for one drive and one set-up, with 5 seeds at 412 particles per cell; the C5 extrapolation is an overcount and not used. We do not claim that HHS is wrong: our heating plateau is half of theirs and the reason is unresolved.

Conclusions. In the HHS resonant case, relativistic detuning explains the collapse of the mean-field response and its timing in deterministic continuum models; an independent relativistic PIC reproduces the HHS electron heating to a factor of about two, while a Newtonian pusher overestimates it by a factor of about two and a half; and the continuum methods developed here resolve the first effect but not the second, whose origin in fluctuations absent from the continuum seeds, possibly at the cell scale, is an open and sharply posed question.

DATA AND CODE AVAILABILITY

All code, records and generated results are in the Dark-SPECTRAX repository [20] (`https://github.com/uwplasma/Dark-SPECTRAX`). The records and study scripts of this paper are those of commit `eb36319` of the `main` branch (relativistic grid records carry `a5aab9f`, relativistic Hermite `0873d10`; nonrelativistic studies `a3ef7a6`); the manuscript, figures and number macros are in `paper/` at the commit that follows (`paper/README.md`). No release or tag has been made. Figures, tables and quoted numbers are generated from the records by `paper/make_figures.py`; the record index below maps each figure and table to its files. The parent solver SPECTRAX [7] was used at the commits stored in the records: `3de6b2b` (finite reservoir and coherence), `4090fbe` (strong-drive study), `35231ab` (collisions) and, for the relativistic Hermite model, `6a6606b` of `integration/dark-baseline-2`. These belong to unmerged integration branches of SPECTRAX that collect open parent pull requests with the features used here (field-scaled closure, Lawson factor, collision conventions, relativistic option and its pump-frame support); they are not releases. The PIC runs use Dark-JAX-in-Cell at `d547579` on JAX-in-Cell at `2d693cb`. The relativistic grid solver uses NumPy and SciPy only. Checks were committed before the runs in `studies/r1_common.py`, `r1nl_gates.py`, `r2_gates.py`, `relgrid_gates.py`, `relherm/relherm_gates.py`, `relhhs_gates.py`,

`relpic/gates.py` and `relpic/conv_gates.py`.

Record index (paths relative to the repository root). Fig. 1 and Table I: `docs/_static/b00/run.json`, `studies/instab/otsi.json`, `studies/c05/run.json`, `studies/gpures/c08.json`, `studies/c09/run.json` (summary in `docs/results.md`). Fig. 2: `studies/highk/eig.json`, `studies/gpures/summary.json`, `studies/rclosure/stagef.json`. Table III, Fig. 3a: `studies/r1nl/stage1_summary.json`, `stage2_summary.json`. Fig. 3b: `studies/r1nl/followup_summary.json`. Fig. 3c and the weak-drive reservoir runs: `studies/r1/coherence2_summary.json`, `studies/r1/reservoir_summary.json`. Table IV, Fig. 4: `studies/r2/summary.json`. Relativistic grid (Sec. VII, Table V): `studies/relgrid/summary.json`, `gates_*.json`, `runs/`. Relativistic Hermite (Sec. II E, VII, IX): `studies/relherm/gates_*.json`, `studies/relhhs/summary.json`, `gates_*.json`, `runs/`. PIC (Sec. VIII, Table VI, Fig. 5): `studies/relpic/conv_summary.json`, `records/`, `digitization/hhs_fig2.json`; nonrelativistic comparison of Table V: `studies/r1nl/runs`; k-range grid: `studies/hkr/summary_grid.json`. Failed and amended checks (Table II): `docs/status.md`, `docs/results.md`.

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